When the Model Doesn't Know It's Wrong: Accuracy and Calibration of Image-Based Photometric Redshift Estimation under a Brightness Shift

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***Abstract*—Photometric redshift (photo-z) estimation from galaxy images underpins modern cosmological surveys, yet the spectroscopic samples available for training are systematically biased toward bright, nearby galaxies, while the scientifically important targets are faint and distant. We treat this train–test mismatch as a controlled distribution-shift problem: models are trained only on bright galaxies and evaluated on a held-out faint population whose redshift distribution differs strongly (Kolmogorov–Smirnov distance 0.72). We propose a multimodal network that fuses five-band imagery with catalog photometry and predicts a full redshift distribution over discretized bins rather than a scalar, trained with a soft-target, imbalance-aware objective. Relative to a single-Gaussian image-only baseline, our model roughly halves the robust scatter on faint galaxies (σ_{MAD}: $0.090 \rightarrow 0.041$) and reduces the catastrophic-outlier fraction from 30% to 19%. A seeded ablation isolates three design axes and shows that the distribution-valued output, not the added modality, is the single largest contributor, while inverse-density reweighting yields no net accuracy gain—a reported negative result. Finally, evaluating the full predictive distributions, we find the model is *underconfident* in-distribution yet *overconfident* out-of-distribution, with the transition aligned to the training brightness limit: its uncertainty estimates are least trustworthy exactly where its predictions are worst.**

***Index Terms*—photometric redshift, distribution shift, uncertainty calibration, convolutional neural networks, multimodal learning, imbalanced regression, astronomy.**

I. Introduction

The redshift z of a galaxy measures how much cosmic expansion has stretched its light, and it serves as a proxy for distance. Precise redshifts come from spectroscopy, which is observationally expensive; as a result, wide surveys depend on *photometric* redshifts inferred from broadband imaging. Deep learning has become competitive with, and often superior to, classical template-fitting in regions where labeled data are abundant [2], [3].

A structural difficulty persists, however. The spectroscopic catalogs used as training labels over-represent bright, low-redshift galaxies, because those are the easiest to confirm spectroscopically. The galaxies that dominate deep imaging surveys are faint and high-redshift. Networks are therefore trained on an easy slice of the population and deployed on a harder one, and prior work repeatedly notes that accuracy degrades on faint, high-z sources and attributes this to data imbalance. Most treatments, though, report this as an incidental limitation rather than studying it directly.

We make the shift the object of study. On the public GalaxiesML dataset [1] we construct a controlled brightness shift: training, validation, and calibration data are drawn exclusively from bright galaxies, while evaluation is performed on a held-out faint population. This isolates the shift and lets us pose two questions precisely: (i) which modeling choices actually recover faint-galaxy *accuracy*, and (ii) does the model's reported *uncertainty* remain honest under the shift?

Contributions.

- A controlled train-on-bright / test-on-faint protocol on public data, with a large, quantified distribution shift (KS $= 0.72$).
- A multimodal, distribution-valued photo-z model that substantially improves faint-galaxy accuracy over a standard image-only regressor.
- A seeded ablation that separates the effect of the output formulation, the photometry modality, and imbalance reweighting, including a negative result for reweighting.
- A full-distribution calibration analysis revealing a two-regime failure mode—underconfidence in-distribution, overconfidence out-of-distribution—tied to the training brightness limit.

II. Related Work

Machine-learning photo-z methods divide broadly into tabular models that operate on catalog magnitudes and colors, and image-based convolutional networks that additionally exploit morphology. Casting redshift estimation as classification over discrete bins, thereby yielding a predictive distribution, follows Pasquet et al. [2]. Multimodal fusion of imagery and photometry has been shown to improve precision across datasets [3], [4]. Imbalanced-regression techniques such as Label Distribution Smoothing [5] reweight or smooth the

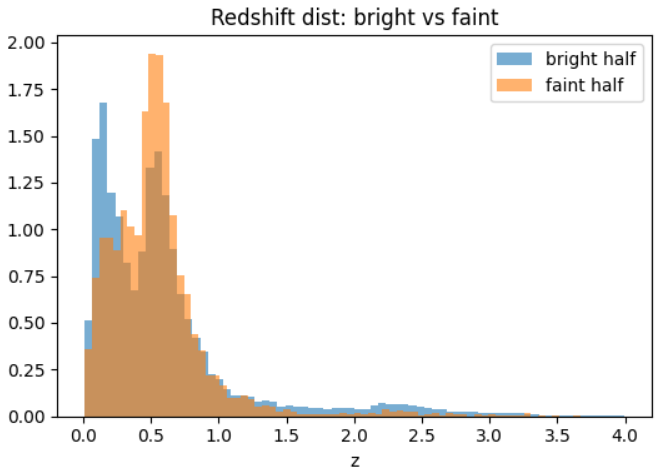


Fig. 1. Distribution of bright and faint images across z-values

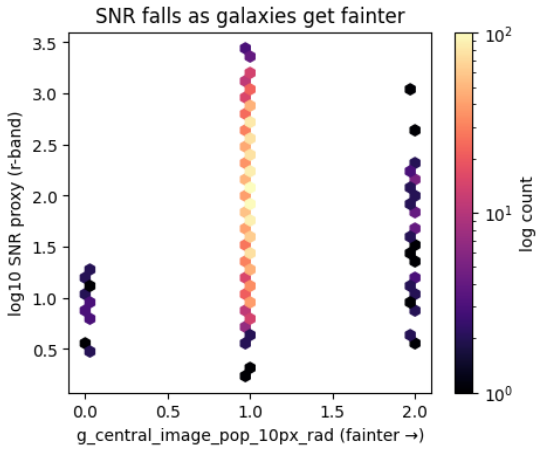


Fig. 2. r-band signal-to-noise proxy versus magnitude. SNR collapses as galaxies fade, explaining why image-only estimation deteriorates in the faint regime.

label density to counter under-represented targets. Our residual backbone uses squeeze-and-excitation channel attention [7] on a residual network [6]. Across this literature, accuracy loss on faint and high-z sources is widely acknowledged, but uncertainty *calibration* under a controlled shift is comparatively unexamined; we target that gap.

III. Data and Exploratory Analysis

We use the $5 \times 64 \times 64$ subset of GalaxiesML [1], which provides g, r, i, z, y image cutouts together with catalog photometry and spectroscopic redshifts. After removing sources with non-finite or non-physical redshift ($z \leq 0$ or $z > 4$), non-finite r-band magnitude, and unmeasured bands (flagged by anomalous magnitude uncertainties), we retain 204,573 galaxies.

Three properties of the data motivate our design. The redshift distribution is heavily skewed toward low z: 93% of galaxies lie below $z = 1.5$ and the median is 0.49, as seen in Figure 1. This skew means a squared-error objective is dominated by the low-z bulk and motivates outlier-aware metrics. Second, brightness and redshift are correlated: fainter galaxies are, on average, more distant. Third, a signal-to-noise (SNR) proxy—the ratio of central peak flux to background noise in the r band—falls sharply with magnitude (Fig. 2), so faint galaxies are harder both because they are farther and because their images carry less information.

IV. Method

A. Input and Output Formulation

Each galaxy is a five-channel image $x_{\text{img}} \in \mathbb{R}^{5\times64\times64}$ (g, r, i, z, y), optionally paired with a photometry vector $x_{\text{ph}} \in \mathbb{R}^9$: the five cmodel magnitudes and four adjacent colors $(g{-}r,\ r{-}i,\ i{-}z,\ z{-}y)$. We do not regress a scalar redshift. We discretize $z \in [0, 4]$ into $B = 200$ uniform bins with centers $\{z_k\}$ and predict a categorical distribution $\mathbf{p} = (p_1, \ldots, p_B)$. The point estimate and predictive uncertainty are the mean and standard deviation of this distribution,

$$\hat{z} = \sum_{k=1}^{B} p_k z_k, \qquad \hat{\sigma} = \Big(\sum_{k=1}^{B} p_k (z_k - \hat{z})^2 \Big)^{1/2}. \tag{1}$$

The distribution-valued output is motivated by the color–redshift degeneracy, which renders $p(z \mid x)$ genuinely multimodal for faint sources: a scalar regressor minimizes expected error by collapsing toward the dominant low-z mode, whereas a distribution can retain competing modes and express ambiguity through $\hat{\sigma}$.

B. Architecture

The model (Fig. 3) has two branches feeding a fusion head. The image branch is a residual network [6] with four stages of widths $32 \to 64 \to 128 \to 256$, reducing spatial resolution $64 \to 4$. Each residual block is two 3×3 convolution–batch-norm–ReLU layers with an identity or projection shortcut, followed by a squeeze-and-excitation (SE) block [7] that applies learned per-channel gating—valuable for emphasizing informative bands when signal is weak. Global average pooling yields a 256-d embedding. The photometry branch is a two-layer MLP $(9 \to 64 \to 64)$. The concatenated 320-d representation is decoded by a fully connected head with dropout into B logits, followed by a softmax.

C. Loss Function

We train with a soft-target cross-entropy. The scalar label z is mapped to a target distribution $\mathbf{t}$ by placing a Gaussian kernel of width $s \approx 1.5$ bins at z and normalizing,

$$t_k = \frac{\exp\big(-(z_k - z)^2 / 2s^2\big)}{\sum_{j=1}^{B} \exp\big(-(z_j - z)^2 / 2s^2\big)}. \tag{2}$$

Unlike a one-hot label, (2) preserves the ordinal structure of redshift—neighboring bins are treated as similar—and regu-

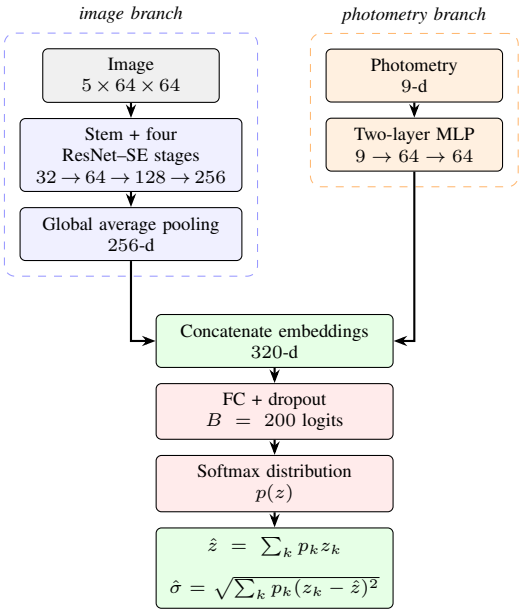


Fig. 3. Multimodal architecture. A residual image branch with squeeze-and-excitation attention yields a 256-d embedding; a photometry MLP yields a 64-d embedding. The fused representation is decoded into a $B = 200$-bin redshift distribution, from which the point estimate $\hat{z}$ and uncertainty $\hat{\sigma}$ are computed via (1).

larizes sparse high-z bins. The per-sample loss is a weighted cross-entropy against this soft target,

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w(z_i)\sum_{k=1}^{B} t_{ik}\,\log p_{ik}. \tag{3}$$

The optional imbalance weight follows Label Distribution Smoothing [5]: with $\tilde{\rho}(z)$ the Gaussian-smoothed empirical training density over bins,

$$w(z) \;\propto\; \min\Big(\tilde{\rho}(z)^{-1/2},\, c\Big), \tag{4}$$

normalized to unit mean over the training set, with cap $c = 10$. The cap is a deliberate choice: uncapped inverse-density weights reached $28\times$ on the sparse $z \to 4$ tail, letting a handful of noisy galaxies destabilize training. We reject plain mean-squared error, which yields a bare point with no uncertainty and lets the low-z bulk dominate the tail.

a) Probabilistic-regression comparators: For the single-Gaussian baseline the head emits a mean and log-variance and is trained with the Gaussian negative log-likelihood

$$\mathcal{L}_{\mathrm{G}} = \frac{1}{N}\sum_{i}\Big[\tfrac{1}{2}\log\sigma_i^2 + \frac{(z_i-\mu_i)^2}{2\sigma_i^2}\Big]. \tag{5}$$

A mixture density network (MDN) generalizes this to K Gaussians, $p(z \mid x) = \sum_k \pi_k \mathcal{N}(z;\mu_k,\sigma_k^2)$, trained by the mixture negative log-likelihood

$$\mathcal{L}_{\mathrm{MDN}} = -\frac{1}{N}\sum_{i}\log\sum_{k=1}^{K}\pi_{ik}\,\mathcal{N}(z_i;\mu_{ik},\sigma_{ik}^2), \tag{6}$$

with point estimate $\hat{z} = \sum_k \pi_k\mu_k$ and, by the law of total variance, $\hat{\sigma}^2 = \sum_k \pi_k(\sigma_k^2+\mu_k^2) - \hat{z}^2$.

D. Optimization and Design Decisions

We use AdamW [8] (weight decay 10^{-4}), an initial learning rate of 3×10^{-4} with cosine annealing [9] over 45 epochs, gradient-norm clipping at 5.0, batch size 256, and dropout 0.3 in the head. Two choices reflect the task rather than convention. First, model selection uses validation σ_{MAD} rather than validation loss, because robust scatter—not the surrogate cross-entropy—is the quantity of interest. Second, images are $\mathrm{arcsinh}$-stretched to compress the large astronomical dynamic range and then per-band standardized using *training statistics only*, avoiding leakage of test information into normalization; random flips and 90° rotations augment training, exploiting the absence of a canonical galaxy orientation.

V. Experimental Setup and Baselines

A. Splits and the induced shift

Brightness is defined by r-band cmodel magnitude, split at the median ($r = 20.31$). Brighter galaxies form the training, validation, calibration, and in-distribution (bright) test pools; all fainter galaxies form the out-of-distribution (OOD) faint-test set, subsampled to 18,000 for a balanced comparison against the 15,342 bright-test galaxies. The induced shift is severe: the median redshift rises from 0.22 (bright) to 0.65 (faint), the fraction with $z > 0.5$ rises from 13.1% to 83.8%, and the Kolmogorov–Smirnov distance between the two redshift distributions is 0.72. The faint set is therefore a genuine OOD challenge, not a mild perturbation.

B. Baselines and fair-comparison controls

The primary baseline is an image-only heteroscedastic CNN trained with (5). Within the proposed family we ablate three configurations sharing one backbone: image-only, +photometry, and full (+LDS). A mixture density network (6) serves as an image-only probabilistic-regression comparator. All models use the identical cleaned training file, the identical shared split indices, the identical per-band normalization statistics, and identical metric definitions and evaluation sets, ensuring that every reported difference is attributable to the modeling change and not to the data pipeline. Complete configurations are averaged over three random seeds.

C. Metrics

With normalized residual $\Delta z = (\hat{z}-z)/(1+z)$ we report bias (mean Δz), robust scatter $\sigma_{\mathrm{MAD}} = 1.4826\cdot \mathrm{median}\,|\Delta z - \mathrm{median}(\Delta z)|$, and catastrophic-outlier fraction $\eta = \frac{1}{N}\sum_i \mathbf{1}[\,|\Delta z_i| > 0.15\,]$. For calibration (Section VII) we use the probability integral transform, central-interval coverage, and the continuous ranked probability score, all computed from the full predicted distribution.

VI. Results and Ablations

Table I reports the main results. Moving from the single-Gaussian baseline to the distribution-valued models roughly halves faint scatter ($0.090 \to 0.047$) and reduces catastrophic outliers from 30% to 22%. We flag that this comparison changes two factors at once—the output head and a stronger

backbone—so it upper-bounds rather than isolates the effect of the output formulation.

Within the shared-backbone ablations the picture is clean. Adding photometry is the single largest lever: faint σ_{MAD} falls from 0.0467 to 0.0414 and faint bias roughly halves ($0.043 \rightarrow 0.025$, the best of any model), consistent with catalog colors supplying information that the low-SNR images cannot. Imbalance reweighting, by contrast, is *not* a net win: relative to +photometry it slightly worsens faint scatter ($0.0414 \rightarrow 0.0432$) and bias, buying only a marginal reduction in the outlier fraction. This matches the known caution that tail-reweighting trades bulk accuracy for tail coverage, and we report it as a negative result rather than omitting it. Bright accuracy stays strong throughout ($\sigma_{\mathrm{MAD}} \approx 0.010$–$0.011$), confirming that gains on the hard regime do not come at the cost of the easy one. A residual positive faint bias persists across all models, reflecting the color–redshift degeneracy that reweighting cannot remove.

TABLE I
FAINT (OOD) AND BRIGHT (IN-DISTRIBUTION) TEST METRICS. LOWER IS BETTER; $\pm$ DENOTES STANDARD DEVIATION OVER SEEDS. BEST PER COLUMN IN BOLD.

Model	Faint σ_{MAD}	Faint η	Bright σ_{MAD}
Heterosc. baseline (img)	0.090	0.301	0.016
Image-only (binned)	0.0467 ± 0.0042	0.217 ± 0.027	0.0100
+ Photometry	$\mathbf{0.0414} \pm 0.0003$	0.193 ± 0.006	0.0100
Full (+ LDS)	0.0432 ± 0.0017	$\mathbf{0.188} \pm 0.017$	0.0112

VII. CALIBRATION ANALYSIS

Accuracy alone does not establish whether the model's uncertainty is trustworthy. We evaluate the full predicted distribution $p_i(\cdot)$ with three diagnostics. The probability integral transform is the predictive CDF evaluated at the truth,

$$\mathrm{PIT}_i = F_i(z_i) = \sum_{k:\, z_k \le z_i} p_{ik}, \tag{7}$$

which is uniform on $[0, 1]$ for a perfectly calibrated model. Central-interval coverage at level c is the fraction of galaxies whose truth falls within the central-c probability interval $[\, q_i(\frac{1-c}{2}),\, q_i(\frac{1+c}{2}) \,]$ of p_i, and should equal c. The continuous ranked probability score,

$$\mathrm{CRPS}_i = \int_{-\infty}^{\infty} \big(F_i(y) - \mathbf{1}[y \ge z_i]\big)^2 \, dy, \tag{8}$$

rewards distributions that are both sharp and calibrated (lower is better).

Figure 4 shows the results computed from the 200-bin distributions. The reliability diagram (left) shows the bright curve bowing *above* the diagonal—the model is underconfident in-distribution, with intervals wider than warranted—while the faint curve falls *below* it, indicating overconfidence out-of-distribution. The coverage-versus-brightness panel (right) shows this transition is monotonic in magnitude and crosses the nominal targets near the training brightness limit: for the faintest galaxies the nominal 68% interval attains only $\approx 28\%$ coverage and the 95% interval $\approx 63\%$. The per-split CRPS (0.029 bright vs. 0.168 faint) confirms that overall predictive quality is several times worse on faint galaxies. The practical implication is direct: the reported error bars are least reliable precisely where predictions are worst, which would silently corrupt any downstream analysis that consumes them.

VIII. QUALITATIVE ANALYSIS AND FAILURE CASES

The dominant failure mode is physical and visible. On faint galaxies the predictions concentrate around $z \approx 0.5$–1.0 and fail to reach true high-z sources, so the prediction-versus-truth relation flattens well below the identity line (Fig. 5). This is the regression-toward-the-common-answer behavior that the distribution-valued output was designed to mitigate; it does so partially, not completely, because the underlying images of faint high-z galaxies carry little information (Fig. 2).

To make the uncertainty behavior concrete, Fig. 6 shows representative galaxies with their predicted distributions: a confident, well-resolved low-z success with a narrow PDF on the truth; an honest faint failure with a wide, multimodal PDF correctly signaling uncertainty; and a catastrophic outlier with a narrow PDF centered on the wrong redshift—a confident mistake, the case where accuracy and honesty diverge and the one most consequential for downstream use.

IX. LIMITATIONS

We state three limitations plainly. First, the ablation grid is partial: the LDS-only configuration was not run and one +photometry seed is missing, owing to compute constraints; the reported trends are nonetheless consistent across available seeds. Second, the baseline-to-binned comparison confounds the output head with a stronger backbone, so the attribution of that specific gain is an upper bound. Third, we diagnose miscalibration but do not apply a correction; a post-hoc recalibration or split-conformal step on the held-out calibration set is a natural and inexpensive extension that would convert the diagnosis into a fix.

X. CONCLUSION

We framed faint-galaxy photometric redshift estimation as a controlled distribution-shift problem and showed that a multimodal, distribution-valued model substantially improves faint accuracy; that the output formulation contributes more than the added modality; that inverse-density reweighting offers no net gain; and that the model's uncertainty is systematically miscalibrated in opposite directions in- and out-of-distribution, with the transition aligned to the training brightness limit. Reporting accuracy and calibration jointly under a realistic shift is, we argue, the more honest way to characterize photo-z models for deep surveys.

REFERENCES

[1] J. Do *et al.*, "GalaxiesML: a dataset of galaxy images, photometry, redshifts, and structural parameters for machine learning," 2024.

[2] J. Pasquet, E. Bertin, M. Treyer, S. Arnouts, and D. Fouchez, "Photometric redshifts from SDSS images using a convolutional neural network," *Astron. Astrophys.*, vol. 621, A26, 2019.

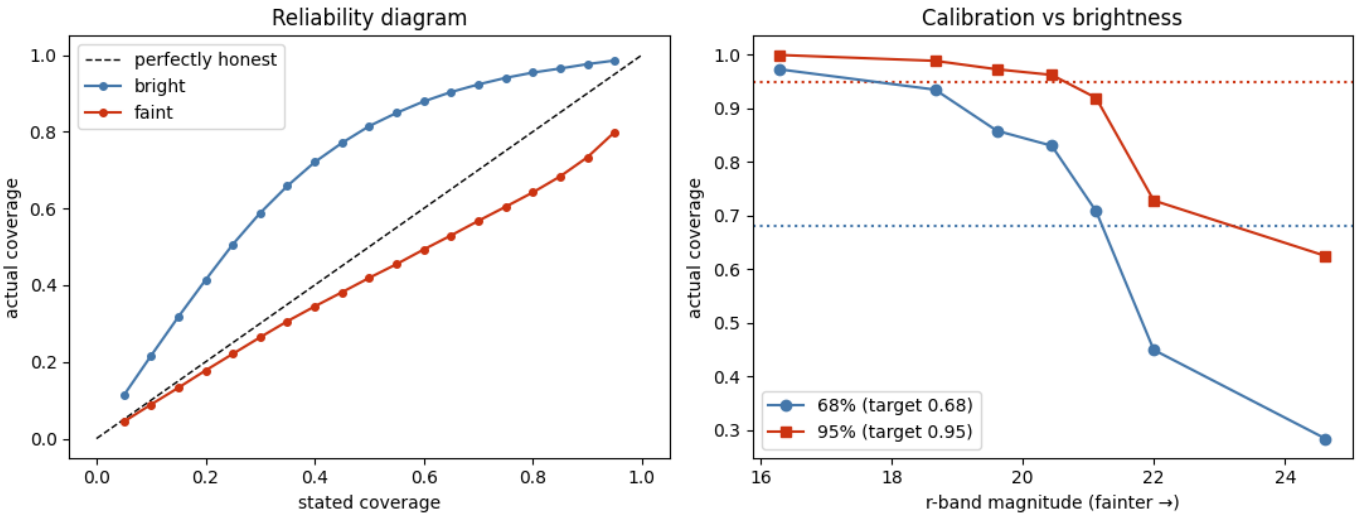


Fig. 4. Calibration from the full predictive distribution. Left: reliability diagram—bright galaxies are underconfident (curve above the diagonal), faint galaxies overconfident (below). Right: actual coverage versus r-band magnitude for the 68% and 95% intervals; dotted lines are nominal targets and coverage degrades monotonically toward the faint end.

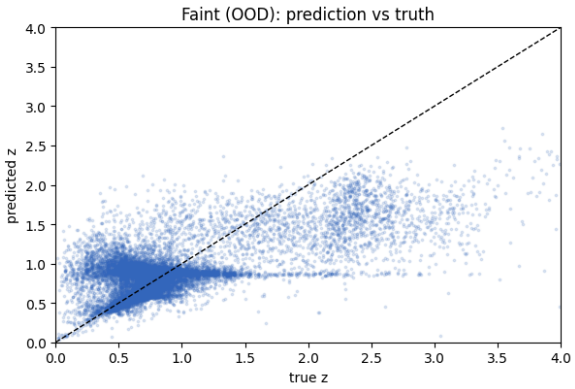


Fig. 5. Predicted versus true redshift on the faint (OOD) set. Predictions saturate below the identity line for high-z galaxies, the characteristic faint-galaxy failure.

[3] M. Treyer *et al.*, "Multimodality for improved CNN photometric redshifts," *Astron. Astrophys.*, 2024.

[4] "Photometric redshift estimation for emission-line galaxies of the DESI Legacy Imaging Surveys by CNN-MLP," 2025.

[5] Y. Yang, K. Zha, Y. Chen, H. Wang, and D. Katabi, "Delving into deep imbalanced regression," in *Proc. ICML*, 2021.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. CVPR*, 2016.

[7] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. CVPR*, 2018.

[8] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[9] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. ICLR*, 2017.

Appendix A
Preliminary Mixture Density Network

As a probabilistic-regression counterpart to the binned model, we additionally trained a mixture density network (MDN) that predicts a continuous redshift distribution as a Gaussian mixture. We report it in an appendix rather than the main text because, owing to compute limits, it was trained only as a reduced-scale run; the numbers below are therefore indicative, not directly comparable to the fully trained models in Table I. Nonetheless, the experiment is informative: even at this scale the MDN independently reproduces the two-regime behavior that is the central finding of the paper.

A. Model and Training

The MDN uses a simple three-block convolutional backbone ($32 \to 64 \to 128$ channels, each followed by max-pooling) feeding a 512-unit fully connected layer with dropout, and three linear heads that emit the mixing weights $\boldsymbol{\pi}$ (softmax), component means $\boldsymbol{\mu}$, and component standard deviations $\boldsymbol{\sigma}$ (softplus), with $K = 3$ Gaussian components. It is image-only (no photometry branch) and is trained by the mixture negative log-likelihood of Eq. (6). The point estimate is the mixture mean $\hat{z} = \sum_k \pi_k \mu_k$, and the predictive uncertainty is the mixture standard deviation obtained via the law of total variance, $\hat{\sigma}^2 = \sum_k \pi_k(\sigma_k^2 + \mu_k^2) - \hat{z}^2$.

Crucially, the MDN uses the *same* shared brightness split and the *same* per-band normalization statistics as the main models, so its train/test populations are identical. The reduced-scale configuration used here is a 2,000-galaxy training subset and a 500-galaxy validation subset, trained for 10 epochs. Validation was monitored by NLL, with the best epoch (val NLL -1.93) retained. On the validation subset the retained model attains a mean absolute error of 0.065 and RMSE 0.207 on the expected-value estimate, with a mean predictive uncertainty of 0.101.

B. Behavior under the Brightness Shift

Table II evaluates the model on the shared combined test set, binned by r-band magnitude. Two trends mirror the main results. First, error grows monotonically with faintness,

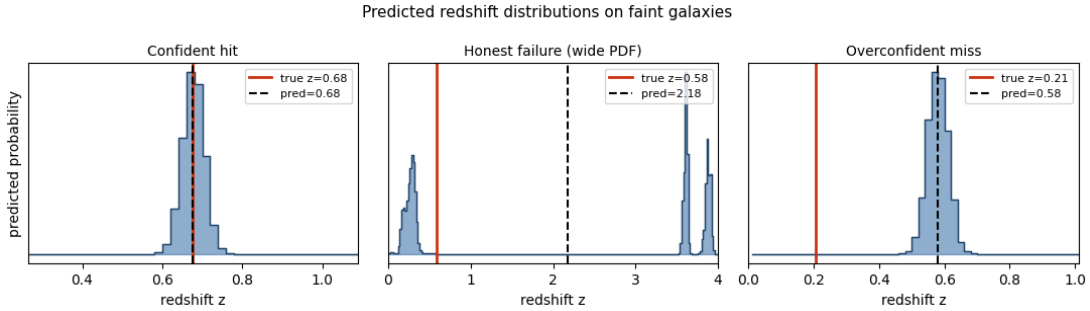


Fig. 6. Representative galaxies with predicted redshift distributions (vertical line: true z). Left: a confident low-z success. Middle: an honest faint failure—wide, multimodal uncertainty. Right: a catastrophic outlier—a narrow PDF centered on the wrong redshift.

from MAE $= 0.020$ for the brightest bin to 0.592 for the faintest—an order of magnitude. Second, and more tellingly, the empirical 2σ coverage stays near or above the nominal 0.95 for the bright bins (indicating slightly conservative, i.e. underconfident, intervals in-distribution) and then collapses to 0.39 for the faintest bin. Notably, the reported uncertainty σ actually *decreases* in the faintest bin ($0.325 \rightarrow 0.218$) even as the error keeps rising, so the model becomes more confident exactly as it becomes more wrong—the overconfidence failure identified in Section VII, here reproduced by an entirely different (mixture-based) uncertainty mechanism.

TABLE II
PRELIMINARY MDN METRICS BY r-BAND MAGNITUDE BIN ON THE SHARED COMBINED TEST SET (REDUCED-SCALE RUN; INDICATIVE ONLY).

Bin (r-mag)	Count	MAE	Mean σ	Coverage (2σ)
14.3–18.4	5557	0.020	0.037	0.991
18.4–19.4	5557	0.056	0.093	0.987
19.4–20.5	5557	0.135	0.198	0.985
20.5–21.2	5557	0.162	0.232	0.983
21.2–22.3	5557	0.366	0.325	0.874
22.3–26.7	5557	0.592	0.218	**0.390**

C. Takeaway

Because it was trained at reduced scale, the MDN's absolute errors are not competitive with the fully trained models and we do not include it in the main comparison. Its value is corroborative: a distinct probabilistic model, trained on the identical shift, exhibits the same qualitative signature— degrading accuracy and collapsing faint-tail coverage—strengthening the paper's conclusion that image-based photo-z uncertainty becomes overconfident out-of-distribution. A full-scale MDN run is left to future work.

APPENDIX B
GRIZY IMAGES

A. What is grizY

In astronomy, grizY (or grizy) refers to a standardized photometric filter system used to capture light across visible and near-infrared wavelengths.

TABLE III
GRIZY SYSTEM SPANS FROM THE GREEN/BLUE OPTICAL SPECTRUM INTO THE NEAR-INFRARED REGION (400 NM TO 1000 NM)

Band	**Spectral Region**	**Central Wavelength ($\approx$)**
g	Green (Visible)	$\sim$480 nm
r	Red (Visible)	$\sim$620 nm
i	Far-Red (Optical/IR edge)	$\sim$750 nm
z	Near-Infrared	
Y	Far Near-Infrared	$\sim$1000 nm (1.0 μm)

Instead of taking a single standard photograph, astronomical surveys map the sky by capturing images through a series of broad optical filters. Each bandpass filter isolates a specific segment of the electromagnetic spectrum, allowing astronomers to measure object brightness, color, temperature, distance (photometric redshift), and chemical composition.

B. Key Applications in Astronomy

1) Composite "True-Color" & False-Color Images: Human eyes cannot process multi-band astronomical data directly. By mapping three of these bands (e.g., $g \rightarrow$ Blue, $r \rightarrow$ Green, $i \rightarrow$ Red), astronomers generate color composite images that highlight structural and stellar differences across galaxies.
2) Photometric Redshifts ($photo{-}z$): As light travels across expanding space, it shifts toward redder wavelengths. Comparing an object's brightness across the g, r, i, z, and Y filters yields its overall spectral energy distribution (SED), allowing astronomers to estimate galaxy distances without needing time-consuming spectroscopy.
3) Dust and Extinction Correction: Cosmic dust scatters shorter blue wavelengths (g) while letting longer near-infrared wavelengths (z, Y) pass through. Comparing fluxes across $grizY$ allows researchers to measure how much dust obscures a star or galaxy.
4) Time-Domain Astronomy & Transients: Wide-field sky surveys re-image the same field across all five filters to detect variable stars, supernovae, asteroids, and gravitational wave optical counterparts.



Fig. 7. Example of five channels from a sample image from the dataset